



Variations on projection of 2-mode network

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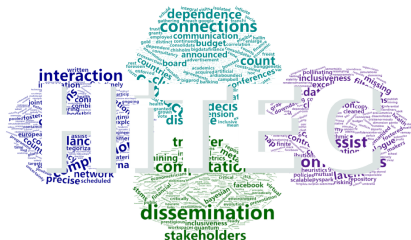
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<https://github.com/bavla/NormNet/tree/main/TwoMode>

Network = Graph + Data

A *network* $\mathcal{N} = (\mathcal{V}, \mathcal{L}, \mathcal{P}, \mathcal{W})$ consists of:

- a *graph* $\mathcal{G} = (\mathcal{V}, \mathcal{L})$, where $\mathcal{V}$ is the set of nodes, $\mathcal{A}$ is the set of arcs, $\mathcal{E}$ is the set of edges, and $\mathcal{L} = \mathcal{E} \cup \mathcal{A}$ is the set of links.
 $n = |\mathcal{V}|$, $m = |\mathcal{L}|$
- $\mathcal{P}$ *node value functions* / properties: $p: \mathcal{V} \rightarrow A$
- $\mathcal{W}$ *link value functions* / weights: $w: \mathcal{L} \rightarrow B$

In a *two-mode* network $\mathcal{N} = ((\mathcal{U}, \mathcal{V}), \mathcal{L}, \mathcal{P}, \mathcal{W})$ the set of nodes consists of two disjoint sets of nodes $\mathcal{U}$ and $\mathcal{V}$, and all the links from $\mathcal{L}$ have one end node in $\mathcal{U}$ and the other in $\mathcal{V}$.

$$\text{wod}(v) = \sum_{e \in \text{ostar}(v)} w(e) \quad \text{and} \quad \text{wid}(v) = \sum_{e \in \text{istar}(v)} w(e)$$

where $\text{ostar}(v)$ is the out-star – links leading out of node v , etc.

Weighted network matrix

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$$\text{wto}(v, u) = \sum_{e \in \text{ostar}(v) \cap \text{istar}(u)} w(e)$$

A two mode network $\mathcal{N} = ((\mathcal{U}, \mathcal{V}), \mathcal{L}, w)$ can be represented with the corresponding **matrix** $\mathbf{UV} = [uv[u, v]]_{u \in \mathcal{U}, v \in \mathcal{V}}$ defined as

$$uv[u, v] = \begin{cases} \text{wto}(u, v) & (u, v) \in \mathcal{L} \\ \square & \text{otherwise} \end{cases}$$

We extend the set of weight values with a new zero element $\square$ (no link) with rules $\square + a = a$ and $\square \cdot a = \square$.

In the case $\mathcal{U} = \mathcal{V}$ we obtain a square matrix of an ordinary network.

A network $\mathcal{N}$ is **binary** iff the weight w is constant $w = 1$.

The **transpose** $\mathbf{UV}^T$ of matrix $\mathbf{UV}$ is a matrix on $\mathcal{U} \times \mathcal{V}$ with entries $uv^T[v, u] = uv[u, v]$. $\mathbf{VU} = \mathbf{UV}^T$

Reversed network $\tilde{\mathcal{N}} = ((\mathcal{V}, \mathcal{U}), \tilde{\mathcal{L}}, vu)$, $\tilde{\mathcal{L}} = \{(v, u) : (u, v) \in \mathcal{L}\}$, $vu(v, u) = uv(u, v)$ for $(u, v) \in \mathcal{L}$.

$$\text{wod}_{VU}(v) = \text{wid}_{UV}(v), \dots$$

In the following, we will often identify networks by their matrices.

The **total weight** of links in the network $\mathcal{N} = (\mathcal{V}, \mathcal{L}, w)$ with matrix $\mathbf{N} = [n[u, v]]$

$$T(\mathbf{N}) = \sum_{(u,v) \in \mathcal{L}} w(u, v) = \sum_{u,v} n[u, v] = \sum_u \text{wod}_N(u) = \sum_v \text{wid}_N(v)$$



Some network transformations

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$S(\mathbf{N})$ transforms a network $\mathbf{N}$ into a simple undirected network with link weights equal to the sum of corresponding arc weights.

$D_0(\mathbf{N})$ removes loops from a network $\mathbf{N}$.

Binarization $\text{bin}(\mathbf{N})$ sets the weight of each link to 1.



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Given a pair of *compatible* networks $\mathcal{N}_A = (\mathcal{I}, \mathcal{K}, \mathcal{A}_A, w_A)$ and $\mathcal{N}_B = (\mathcal{K}, \mathcal{J}, \mathcal{A}_B, w_B)$ with corresponding matrices $\mathbf{A}_{\mathcal{I} \times \mathcal{K}}$ and $\mathbf{B}_{\mathcal{K} \times \mathcal{J}}$ we call a *product of networks* $\mathcal{N}_A$ and $\mathcal{N}_B$ a network $\mathcal{N}_C = (\mathcal{I}, \mathcal{J}, \mathcal{A}_C, w_C)$, where $\mathcal{A}_C = \{(i, j) : i \in \mathcal{I}, j \in \mathcal{J}, c_{i,j} \neq 0\}$ and $w_C(i, j) = c_{i,j}$ for $(i, j) \in \mathcal{A}_C$. The product matrix $\mathbf{C} = [c_{i,j}]_{\mathcal{I} \times \mathcal{J}} = \mathbf{A} * \mathbf{B}$ is defined in the standard way

$$c_{i,j} = \sum_{k \in \mathcal{K}} a_{i,k} \cdot b_{k,j}$$

In the case when $\mathcal{I} = \mathcal{K} = \mathcal{J}$ we are dealing with ordinary one-mode networks (with square matrices).

This definition can be extended to any semiring $(S, +, \cdot, 0, 1)$ [1].

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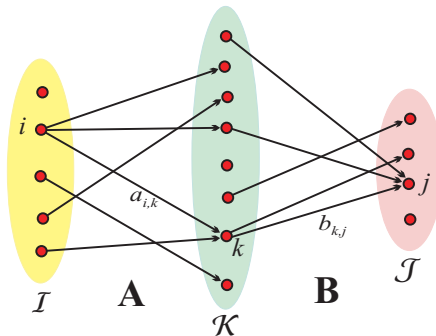
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$$c_{i,j} = \sum_{k \in oN_A(i) \cap iN_B(j)} a_{i,k} \cdot b_{k,j}$$

If both networks $\mathcal{N}_A$ and $\mathcal{N}_B$ are binary the value of $c_{i,j}$ counts the number of ways we can go from $i \in \mathcal{I}$ to $j \in \mathcal{J}$ passing through $\mathcal{K}$, $c_{i,j} = |oN_A(i) \cap iN_B(j)|$. oN are out-neighbors, etc.



Derived networks

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We can use multiplication to obtain new networks from existing *compatible* two-mode networks. For example, from basic bibliographic networks **WA** (Works, Authors) and **WK** we get

$$\mathbf{AK} = \mathbf{WA}^T \cdot \mathbf{WK}$$

a network relating authors to keywords used in their works, and

$$\mathbf{ACiA} = \mathbf{WA}^T \cdot \mathbf{Ci} \cdot \mathbf{WA}$$

is a network of citations between authors [5].

Networks obtained from existing networks using some operations are called *derived* networks. They are very important in the analysis of collections of *linked* networks.

The weight $AK[a, k]$ is equal to the number of times the author a used the keyword k in his/her works.

The weight $ACiA[a, b]$ counts the number of times a work authored by the author a cites a work authored by the author b ; or, more simply, how many times the author a cited the author b .



... and projections

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Two-mode matrices $\mathbf{UV}$ and $\mathbf{UV}^T$ are always compatible on both sides. The corresponding derived one-mode matrices are called *projections*

- *row* projection $\mathbf{VV} = \text{row}(\mathbf{UV}) = \mathbf{UV}^T \cdot \mathbf{UV}$
- *column* projection $\mathbf{UU} = \text{col}(\mathbf{UV}) = \mathbf{UV} \cdot \mathbf{UV}^T$

They are related $\text{row}(\mathbf{UV}) = \text{col}(\mathbf{UV}^T) = \text{col}(\mathbf{VU})$.
Therefore, it is enough to study the properties of one of them.



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For vectors $x = [x_1, x_2, \dots, x_n]$ and $y = [y_1, y_2, \dots, y_m]$ their *outer product* $x \circ y$ is defined as a matrix

$$x \circ y = [x_i \cdot y_j]_{n \times m}$$

then we can express the product $\mathbf{C}$ of two compatible matrices $\mathbf{A}$ and $\mathbf{B}$ as the *outer product decomposition*

$$\mathbf{C} = \mathbf{A} \cdot \mathbf{B} = \sum_k \mathbf{H}_k \quad \text{where} \quad \mathbf{H}_k = \mathbf{A}[:, k] \circ \mathbf{B}[k, :],$$

$\mathbf{A}[:, k]$ is the k -th column of matrix $\mathbf{A}$, and $\mathbf{B}[k, :]$ is the k -th row of matrix $\mathbf{B}$.

$$T(x \circ y) = S_x \cdot S_y \quad \text{where} \quad S_x = \sum_i x_i$$



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In general, each term $\mathbf{H}_k$ in the outer product decomposition has a different value $T(\mathbf{H}_k) = \text{wid}_A(k) \cdot \text{wod}_B(k)$ leading to over-representation of nodes with large values.

$$T(\mathbf{C}) = T\left(\sum_k \mathbf{H}_k\right) = \sum_k T(\mathbf{H}_k) = \sum_k \text{wid}_A(k) \cdot \text{wod}_B(k)$$

To make the contributions of all nodes equal, we can apply the *fractional* approach by normalizing the weights: setting $x' = x/S_x$, we get $S_{x'} = 1$ and therefore $T(x' \circ y') = 1$.

Let $\mathbf{Co} = \text{row}(\mathbf{UV})$ – the *co-occurrence* network.

$$\text{Co}[v, z] = \text{Co}[z, v] \quad T(\mathbf{H}_u) = \text{wod}(u)^2$$

$$T(\mathbf{Co}) = \sum_{u \in \mathcal{U}} T(\mathbf{H}_u) = \sum_{u \in \mathcal{U}} \text{wod}(u)^2$$

Stochastic, Markov: $\hat{\mathcal{U}} = \{u \in \mathcal{U} : \text{wod}_{UV}(u) > 0\}$

$$n(UV)[u, v] = \begin{cases} \frac{uv[u, v]}{\text{wod}_{UV}(u)} & u \in \hat{\mathcal{U}} \\ \square & u \notin \hat{\mathcal{U}} \end{cases} \quad \text{wod}_{n(UV)}(u) = \begin{cases} 1 & u \in \hat{\mathcal{U}} \\ 0 & u \notin \hat{\mathcal{U}} \end{cases}$$

$$T(n(\mathbf{UV})) = \sum_{u \in U} \text{wod}_{n(UV)}(u)^2 = \sum_{u \in \hat{\mathcal{U}}} 1 = |\hat{\mathcal{U}}|$$

Strict, Newman: loops excluded

$$\text{wod}'_{UV}(u) = \sum_{v \neq u} uv[u, v] = \text{wod}_{UV}(u) - uv[u, u]$$

$$U' = \{u \in \mathcal{U} : \text{wod}'_{UV}(u) > 0\}$$

$$s(UV)[u, v] = \begin{cases} \frac{uv[u, v]}{\text{wod}'_{UV}(u)} & u \in \mathcal{U}' \\ \square & u \notin \mathcal{U}' \end{cases} \quad \text{wod}_{s(UV)}(u) = \begin{cases} 1 & u \in \mathcal{U}' \\ 0 & u \notin \mathcal{U}' \end{cases}$$

$$T(s(\mathbf{UV})) = \sum_{u \in U} \text{wod}_{s(UV)}(u)^2 = \sum_{u \in \mathcal{U}'} 1 = |\mathcal{U}'|$$

Weight-preserving normalization

$$p(UV)[u, v] = \begin{cases} \frac{uv[u, v]}{\sqrt{\text{wod}_{UV}(u)}} & u \in \hat{U} \\ \square & u \notin \hat{U} \end{cases}$$

$$\text{wod}_{p(UV)}(u) = \begin{cases} \sqrt{\text{wod}_{UV}(u)} & u \in \hat{U} \\ 0 & u \notin \hat{U} \end{cases}$$

$$T(u) = \text{wod}_{p(UV)}(u)^2 = \text{wod}_{UV}(u)$$

[2]

Fractional projection

$$\mathbf{Cn}_N = S(n(\mathbf{N})^T \cdot n(\mathbf{N}))$$

$$\mathbf{Cn}_N^T = \mathbf{Cn}_N \text{ and } T(\mathbf{Cn}_N) = |\hat{\mathcal{U}}|.$$

Strict fractional projection

$$\mathbf{Cs}_N = S(D_0(s(\mathbf{N})^T \cdot n(\mathbf{N})))$$

$$\mathbf{Cs}_N^T = \mathbf{Cs}_N \text{ and } T(\mathbf{Cs}_N) = |\mathcal{U}'|.$$

Weight-preserving projection

$$\mathbf{Cp}_N = S(p(\mathbf{N})^T \cdot p(\mathbf{N}))$$

$\mathbf{Cp}_N^T = \mathbf{Cp}_N$ and $T(\mathbf{Cp}_N) = T(\mathbf{N})$. The total of weights in the original two-mode network is distributed over arcs of the projection.



Projections through an intermediate relation

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Assume that a network $\mathbf{R}$ on $\mathcal{U} \times \mathcal{U}$ is given. The network

$$\mathbf{Q} = n(\mathbf{UV})^T \cdot \mathbf{R} \cdot n(\mathbf{UV})$$

links nodes from $\mathcal{V}$ *through* the network $\mathbf{R}$.

It holds

- If $\mathbf{R}$ is symmetric, $\mathbf{R}^T = \mathbf{R}$, then also $\mathbf{Q}$ is symmetric, $\mathbf{Q}^T = \mathbf{Q}$.
- $T(\mathbf{Q}) = \sum_{u \in \hat{\mathcal{U}}} \sum_{t \in \hat{\mathcal{U}}} r[u, t] = \hat{T}(\mathbf{R})$.

Therefore, if $\hat{\mathcal{U}} = \mathcal{U}$ then $T(\mathbf{Q}) = T(\mathbf{R})$. The total weight of $\mathbf{R}$ is redistributed over the links of $\mathbf{Q}$.



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Let $c : \mathcal{U} \rightarrow \mathbb{R}$ be a property of nodes. Setting $\mathbf{R} = \text{diag}(c)$ we get

$$T(\mathbf{Q}) = \sum_{u \in \hat{\mathcal{U}}} c(u)$$

The contribution of each active node $u \in \hat{\mathcal{U}}$ to the weights of $\mathbf{Q}$ is $c(u)$.

For example, for the bibliographic network **WA** the property c can be

- the impact factor of the journal in which the work u was published
- the number of references to the work u
- the size (number of characters or words) of the work u



Fractional citations between authors

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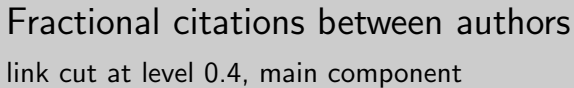
References

From the **OpenAlex** bibliographic database, we obtained in July 2026 a collection of networks (**Ci**, **WA**, **WJ**, **WK**, **WC**) based on works with the keyword `social-network` or `social-network-analysis` and the works cited from them – $|W| = 956445$ and $|A| = 159307$. We computed the network of *fractional citations between authors*

$$\mathbf{ACiAn} = D_0(n(\mathbf{WA})^T \cdot n(\mathbf{Ci}) \cdot n(\mathbf{WA}))$$

The network **ACiAn** is a projection through the normalized citation network $n(\mathbf{Ci})$. Therefore $T(\mathbf{ACiAn}) = T(n(\mathbf{Ci}) = |\hat{\mathcal{W}}|$. Each active work contributes value 1 that is distributed over the weights of the network **ACiAn**.

We made a link cut at level 0.4 (component size distribution: 156: 1, 8: 2, 7: 1, 5: 1, 4: 3, 3: 36, 2: 57). We present the main component.



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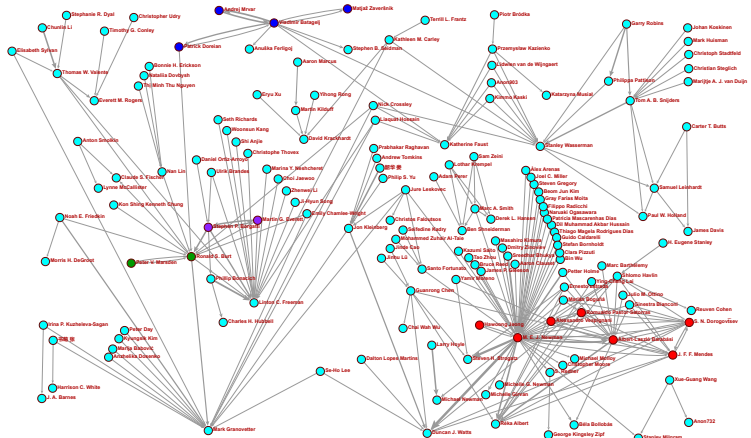
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almost acyclic – small strong components



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- The obtained projections can be analyzed using standard network analysis methods [3, 7, 6].
- This is a traditional recipe for analyzing two-mode networks. Often, the weights are not considered in the analysis, and when they are, we have to be very careful about their meaning. In cases where the direct fractional approach is not meaningful, we can apply the approach proposed for normalized bibliographic coupling in [4, pp. 631–632].
- Traditional projections were defined for binary networks. We extended the definition to weighted networks and proposed some additional options.



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The computational work reported in this presentation was performed using R and the program for analysis and visualization of large networks, Pajek [8]. The code and data are available at [GitHub/bavla](#) [9].

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